



# MBA IGNITE : CAT + OMETs

## Lecture 02 : Logical Reasoning

Arrangements (Part-2)  
Circular Arrangements

**GOURAV GUPTA**



# Recap of Previous Lecture

Linear Arrangements

→ Rules

→ 3 sets

——  $\overline{\downarrow \uparrow}$

→ 1 set → (HW)



# Today's Agenda

1

Basics of Circular Arrangements ✓

2

Questions based on Circular Arrangements ✓



# Homework



## Set -3 (Single row-same direction)

Read the following passage and answer the given questions.



Eight people are seated in a linear row facing north. Anil is seated fourth to the left of Radha. Radha is not seated at the end of the row. Jaggu is an immediate neighbour of Anil. Two people are seated between Parul and Jaggu. Anil is seated to the left of Parul. Nisha is seated to the immediate left of Ekta. Shyam is seated to the left of Manju.

J A S P M R N E .



#Q. How many people are seated to the left of Parul ?

- (A) 1                      (B) 2                      (C) 3                      (D) 4

#Q. The number of people seated between Radha and Ekta is the same as the number of people seated between \_\_\_\_\_ and Parul.

- (A) Shyam              (B) Anil              (C) Jaggu              (D) Nisha

#Q. The number of people seated to the left of Shyam is the same as the number of people seated to the right of \_\_\_\_\_.

- (A) Nisha              (B) Parul              (C) Radha              (D) Jaggu

#Q. Which of these statements is true ?

- (A) Jaggu is seated at the left end of the row  
(B) Four people are seated between Shyam and Ekta  
(C) Parul is an immediate neighbor of Manju  
(D) All of these

#Q. What is the position of Nisha with respect to Parul ?

- (A) 2nd to the left      (B) 3rd to the right  
(C) 2nd to the right      (D) 3rd to the left

## Analysis



J A S P M R N E ✓

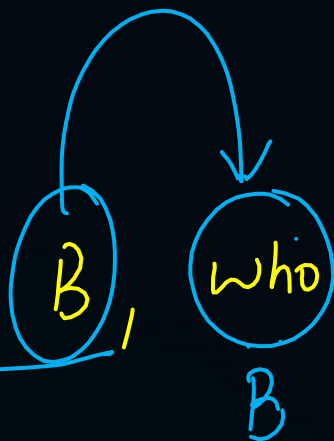
P A J     R   ✗

N E

S M

Eight people are seated in a linear row facing north. Anil is seated fourth to the left of Radha. Radha is not seated at the end of the row. Jaggu is an immediate neighbour of Anil. Two people are seated between Parul and Jaggu. Anil is seated to the left of Parul. Nisha is seated to the immediate left of Ekta. Shyam is seated to the left of Manju.

A is to the left of B, who is at extreme Right



— — — — —



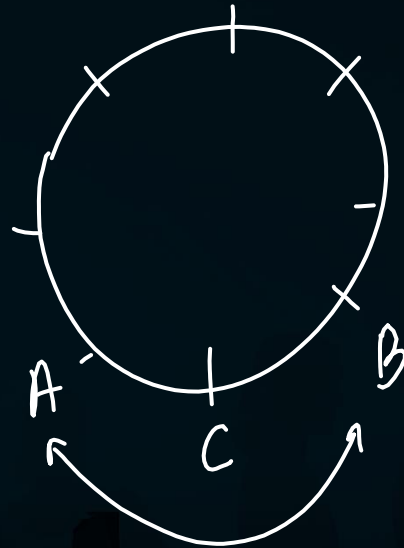
# Basics of Arrangements



# Circular Arrangements

- C sits between A and B
- Only C sits between A and B

} Same.





# Circular Arrangements

- B sits to the left of A
  - B sits to the immediate left of A
- } Same.



# Circular Arrangements

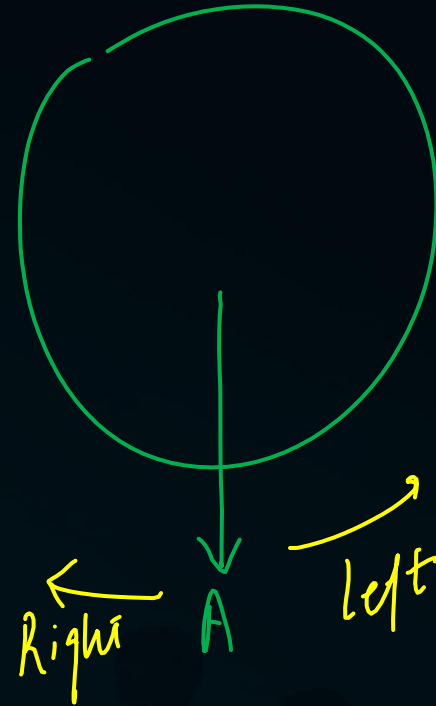
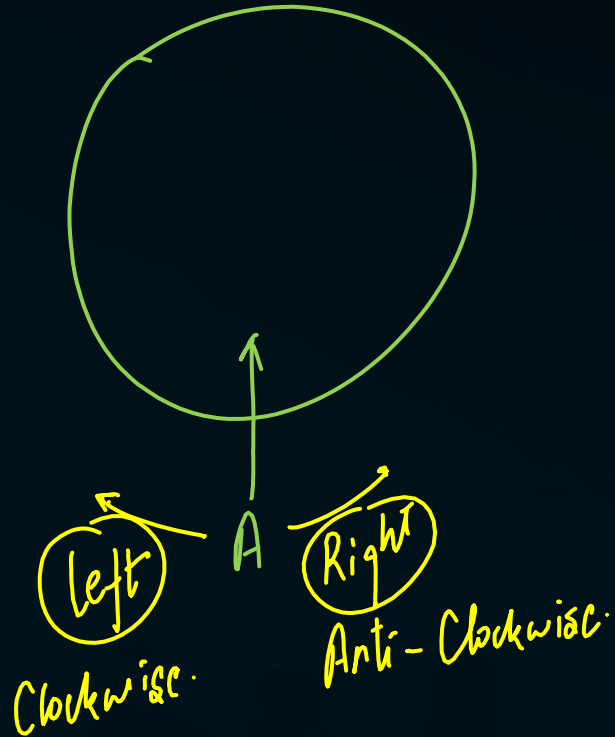


- ① Inside or towards the center
- ② Outside or away from center.

Not Given → Assume → Inside



# Circular Arrangements





# Circular Arrangements

Even



opposite pairs.

Odd.



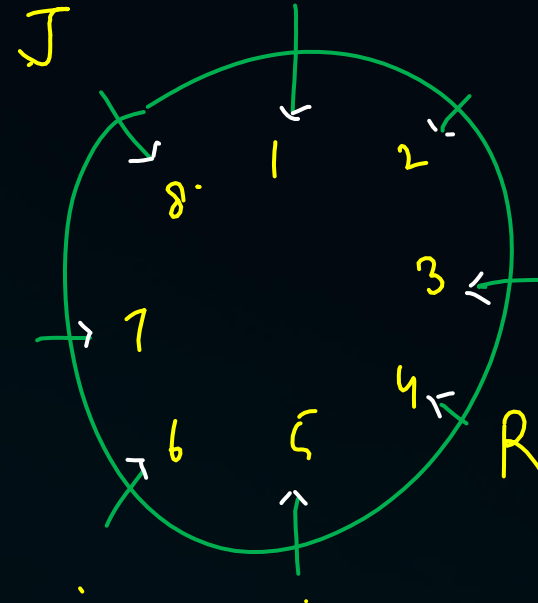
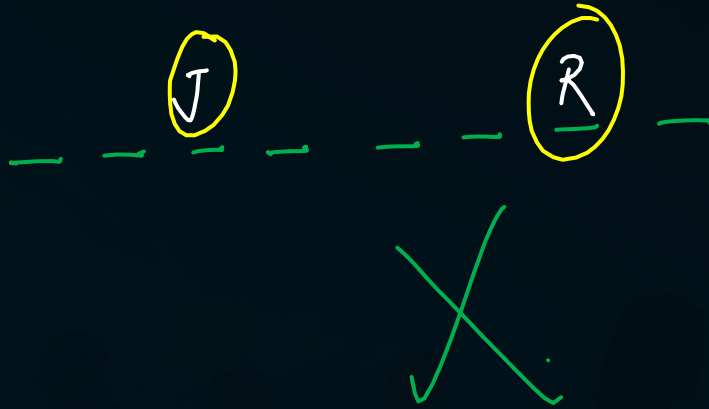
No opposite pairs.

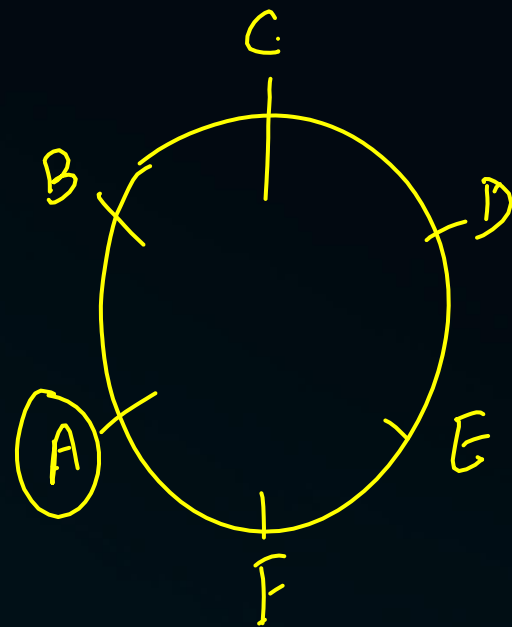
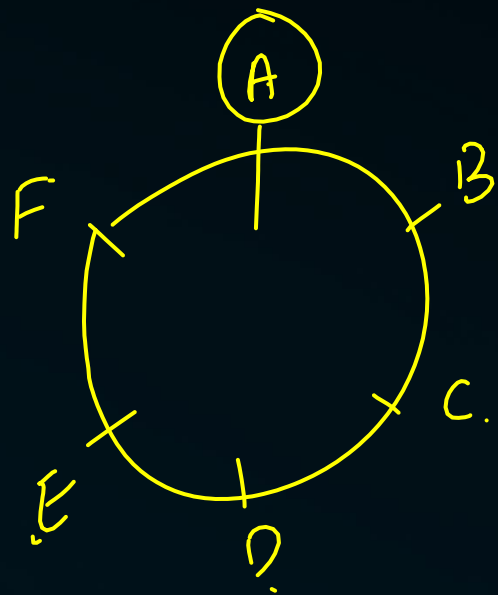


# Circular Arrangements

8 chairs:

(J) is 4<sup>th</sup> to the left of (R)







## **Set -1 : Same Direction**

Refer to the data below and answer the questions that follow.



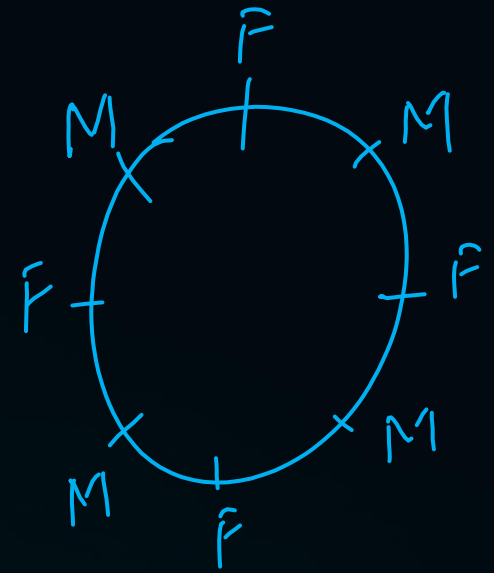
① Arrangement    ② Gender    ③ Couples.

Four married couples are sitting around a round table such that neither any couple nor two men are sitting together. All of them are facing towards the center of the table. The names are A, B, C, D, E, F, G and H.

- A is a male sitting opposite to G.
- E is sitting between F and G.
- To the immediate right of B is H.
- C and H are of same sex and are not sitting opposite to each other.
- Only two persons are sitting between C and B who are married to each other.

Male

Female





#Q. Who is married to D?

- 1) A                      2) B                      ~~3) F~~                      4) G

#Q. Which of following cannot be the married couple according to given conditions?

- 1) G and H ~~x~~                      2) E and A ~~x~~                      ~~3) G and E~~                      4) None of these

#Q. Who is married to the male sitting between H and C?

- 1) C                      2) H                      3) D                      ~~4) E~~

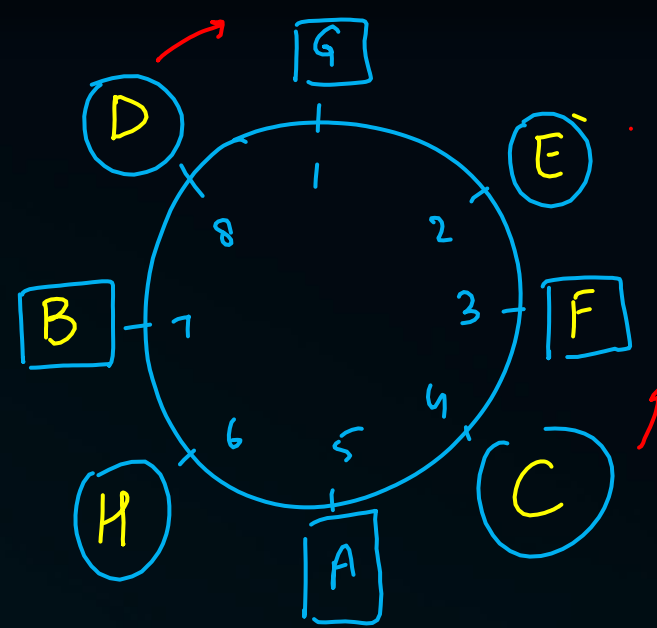
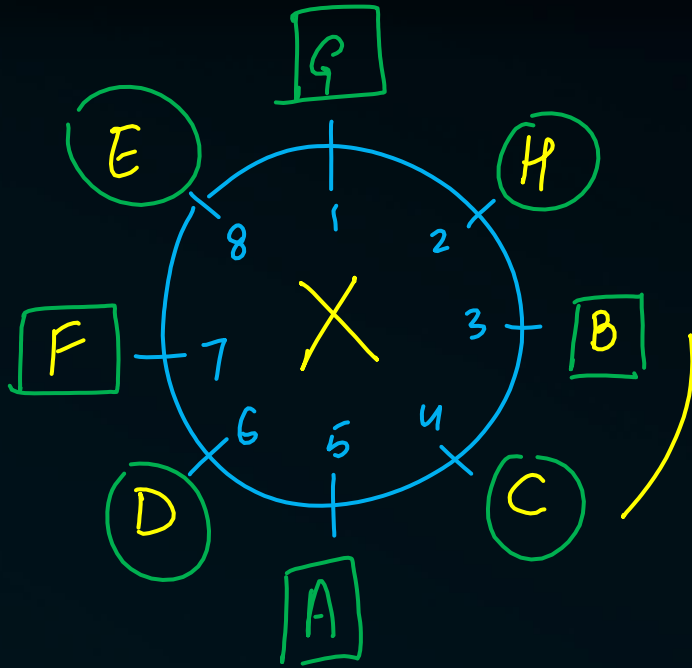
#Q. Who is sitting to the immediate left of the person who is married to F?

- 1) C                      2) H                      3) D                      ~~4) G~~

#Q. Who is married to the person who is sitting second to the right of C?

- 1) A ~~✓~~                      2) H                      3) G                      4) E

# Analysis



Males → A, G, F, B

Females → E, C, H, D

B - C

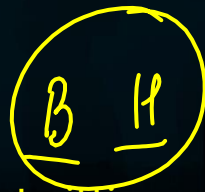
A - E

G - H

F - D

Four married couples are sitting around a round table such that neither any couple nor two men are sitting together. All of them are facing towards the center of the table.

- A is a male sitting opposite to G. ✓
- E is sitting between F and G. ✓
- To the immediate right of B is H. //
- C and H are of same sex and are not sitting opposite to each other. //
- Only two persons are sitting between C and B who are married to each other.





## **Set -2 : Different Directions**

**Read the following passage and answer the given questions.**



Eight friends, A, B, C, D, E, F, G, and H, are sitting around a circular table, not necessarily in the same order. Four of them face away from the centre and four of them face towards the centre.

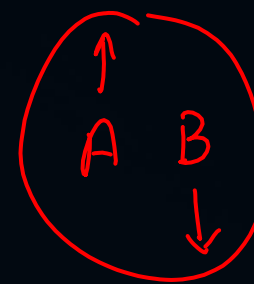
The following information is also known.

- a. Both the immediate neighbours of A, who is not an immediate neighbour of D, face away from the centre.
- b. F faces away and both the immediate neighbours of F face towards the centre.
- c. E sits second to the right of F, while D sits third to the left of F.
- d. G is an immediate neighbour of A, who faces towards the centre.
- e. B, who faces away from the centre, is an immediate neighbour of H, who sits second to the left of C.
- f. D sits third to the right of E.



#Q. How many pairs of friends are there who sit to the right of each other?

- (A) 1 (B) 2 (C) 3 ~~(D) 4~~

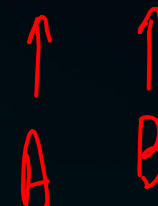


#Q. How many people are seated between B and C when counted clockwise from C?

- (A) 1 ~~(B) 2~~ (C) 3 (D) 4

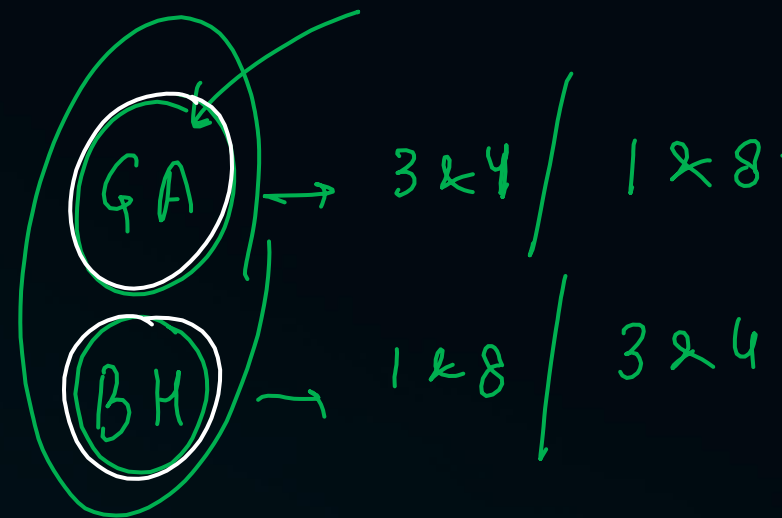
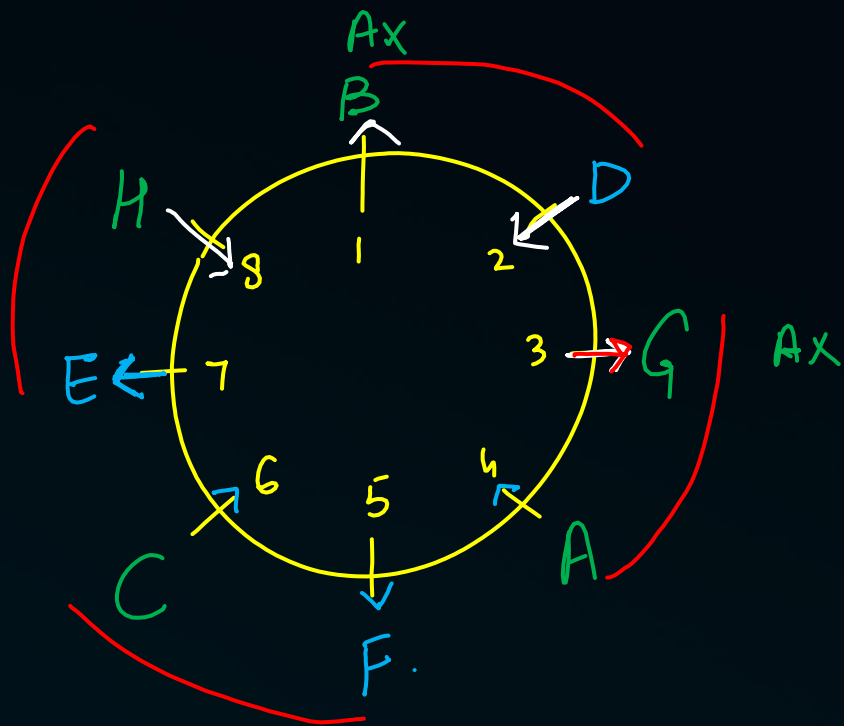
#Q. Who among the following sits third to the right of C?

- (A) A (B) F ~~(C) G~~ (D) C



#Q. Who among the following sits second to the right of E?

- ~~(A) B~~ (B) D (C) C (D) F



Eight friends, A, B, C, D, E, F, G, and H, are sitting around a circular table, not necessarily in the same order. Four of them face away from the centre and four of them face towards the centre.

a. Both the immediate neighbours of A, who is not an immediate neighbour of D, face away from the centre.

b. F faces away and both the immediate neighbours of F face towards the centre.

c. E sits second to the right of F, while D sits third to the left of F.

d. G is an immediate neighbour of A, who faces towards the centre.

e. B, who faces away from the centre, is an immediate neighbour of H, who sits second to the left of C.

f. D sits third to the right of E.



THANK  
*You*

